

Challenge (Habanero)

- Q1.** A gaming company has two types of subscriptions: x for casual gamers and y for pro gamers. If the total revenue from casual gamers is 1200 and from pro gamers is 1800, and the cost per casual gamer is 5 and per pro gamer is 10, the system of equations representing this situation is:
(A) $5x + 10y = 1200$ and $x + y = 1800$
(B) $5x + 10y = 3000$ and $x + y = 300$
(C) $5x + 10y = 1200$ and $5x + 10y = 1800$
(D) $5x = 1200$ and $10y = 1800$
(E) $x + y = 1200$ and $5x + 10y = 3000$
- Q2.** A data analyst is tracking the number of x new users and y active users of a game. If the total number of users is 1000, with 20 new users and 50 active users daily, the equation representing this is:
(A) $x + y = 1000$
(B) $x + y = 70$
(C) $20x + 50y = 1000$
(D) $20x + 50y = 70$
(E) $x - y = 30$
- Q3.** The cost of developing a new game is given by the system of equations: $2x + 5y = 1200$ and $x - 2y = -200$, where x is the cost of development and y is the cost of marketing. Solving this system yields:
(A) $x = 200, y = 100$
(B) $x = 400, y = 100$
(C) $x = 200, y = 200$
(D) $x = 100, y = 200$
(E) $x = 300, y = 100$
- Q4.** In a gaming tournament, x players are in the beginner bracket and y players are in the advanced bracket. If each beginner player pays 20 and each advanced player pays 50, and the total revenue is 2000, the equation is:
(A) $20x + 50y = 2000$
(B) $20x - 50y = 2000$
(C) $x + y = 2000$
(D) $x - y = 2000$
(E) $50x + 20y = 2000$
- Q5.** The score of a game can be represented by the equations $2x - 3y = 10$ and $x + 2y = 8$, where x is the score of team A and y is the score of team B. The solution to this system of equations is:
(A) $x = 2, y = 3$
(B) $x = 4, y = 2$
(C) $x = 6, y = 1$
(D) $x = 1, y = 3$
(E) $x = 2, y = 2$
- Q6.** A game developer has 240 hours to develop a new game and 120 hours to test it. If the development time per level is x hours and the testing time per level is y hours, and there are 10 levels, the system of equations is:
(A) $10x + 10y = 360$ and $x + y = 12$
(B) $10x + 10y = 240$ and $10x + 10y = 120$
(C) $10x = 240$ and $10y = 120$
(D) $10x + 10y = 360$
(E) $x + y = 36$
- Q7.** A tech company has 1500 to spend on x software packages at 20 each and y hardware packages at 50 each. The system of equations representing this situation is:
(A) $20x + 50y = 1500$ and $x + y = 30$
(B) $20x + 50y = 1500$
(C) $20x = 1500$ and $50y = 1500$
(D) $20x + 50y = 1500$ and $x - y = 0$
(E) $x + y = 30$
- Q8.** A gamer has 4 hours to play two games, x and y , where x takes 30 minutes to play and y takes 45 minutes to play. The system of equations for this is:
(A) $2x + 3y = 8$ and $x + y = 4$
(B) $2x + 3y = 8$
(C) $x + y = 8$
(D) $x + 1.5y = 4$ and $x + y = 4$
(E) $0.5x + 0.75y = 4$

Open Ended Questions (FRQ)

- Q9.** A coding logic puzzle has the following conditions: $x + 2y = 10$ and $3x - 2y = 5$. Solve the system of equations and interpret the solution in context. Show all work and explain your answer.
- Q10.** In a scoreboard system, the number of x wins and y losses are related by the equations $2x + 3y = 25$ and $x - 2y = -5$. Evaluate the reasonableness of the solution and provide an interpretation in the context of the scoreboard.

Chef's Tips

- Tip 1: Emphasize the importance of understanding the context of word problems.
- Tip 2: Make sure students can translate real-world problems into systems of equations.
- Tip 3: Encourage students to evaluate the reasonableness of their solutions.